

Empirical Analysis of Copyrights: Evidence from U.S. Multinationals

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The statistical analysis of firm level data on U.S. multinational companies and their foreign affiliates was conducted at the Bureau of Economic Analysis, U.S. Department of Commerce, under arrangements that maintain legal confidentiality requirements. Views expressed in this paper are those of the authors and do not necessarily reflect the official positions of the U.S. Department of Commerce. Presentation is based on joint work with Olena Ivus, Queen's University, Canada.

Overview

Objectives:

- Provide an **example** of an empirical study of IPRs
- Show the **tools** of research used: model/conceptualization, data, methods, & software
- Demonstrate that **IPRs** are broad and connected

This particular study shows:

- Through international licensing, the copyright industry is a leading contributor of global knowledge transfers (more than the manufacturing industry)
- Exploits firm-level data on U.S. MNCs – parents, affiliates, and unaffiliates – across 69 countries, using an index of copyright protection
- Stronger copyright protection limits imitation and infringement, but also raises transfer and adaptation costs.
- The copyright regime influences entry into host markets and raises the volume of transfers, as long as the demand for creative works is sufficiently high; else, market power prevails.
- The above finding varies by industry or sector.
- Effects of copyright reform extend beyond traditional copyright industries to complementary industries

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Motivation

Relevance to Productivity and Economic Development?

- Art Works, Films, Books, Sound Recordings contribute to the stock of knowledge
- Software, Databases, Journals used in scientific research, R&D
- Demand spillovers for Complementary Industries

Their Significance

- Copyright-based industries account for a significant share of GDP, trade, employment
 - cf. World Intellectual Property Office (WIPO) studies on the contribution of copyright-based industries
 - Caution: Non-causal analyses
- Copyright-based industries account for the majority of U.S. outward global licensing
 - Ahead of manufacturing industries: 46% versus 38% during 2010-20

State of Research

Basic Theory (Watt (2000), Landes & Posner (2003), Maskus (2012)):

- Stronger copyright protection and enforcement increases cost of reproduction, reduces imitation, increases return to rights holder, and generates incentives for creation and product launches
- But it increases costs of access and creates transactions costs (burden of clearing rights and/or working around existing rights, especially if limited modes of expression exist)

Empirical Evidence (few):

- Single country (US or Canada)
 - Baker and Cunningham (2006), Ku et al. (2009)
- Single product (books, music, film)
 - Liebowitz (2008), Heald (2009), Png and Wang (2009), Li et al. (2018), Giorcelli and Moser (2020)
- Piracy, File-sharing studies find crowding out is not 1:1
 - Rob and Waldfogel (2006), Oberholzer-Gee and Strumpf (2007), Danaher et al. (2013)
- Limited investigations beyond copyright industry
 - P. Smith et al. (2009) study effect on relevant infrastructure (servers, bandwidth, PCs)
 - Lerner (2011) studies effect on cloud computing

Our Contribution

Goals:

- Estimate relationship between copyright strength and technology transfer of creative works via licensing
- Estimate the degree to which copyright reforms spillover to complementary industries (beyond the creative industries) and enhance their investments

Novelty:

- Uses firm-level data of U.S. MNCs
 - Condition on differential industrial sensitivity to copyright protection (i.e., software may respond differently than films, etc.)
 - Take into account selection bias (i.e., previous studies assume firm is already operating in the local market)
 - Incorporate cost of adaptation of creative works in different host countries (i.e., costs of foreign entry)
 - Analyze affiliated licensing and arms-length licensing (it matters if you're the local market!)

Key findings:

- Copyrights have both *market expansion* and *market power effects* – in theory and in practice.
- They affect the *likelihood* of transfers, with a bias towards intra-firm licensing. Reason: copyrights affect cost of adaptation, which are more onerous for unaffiliated licensing.
- Copyright reforms spill over to supporting industries that produce *inputs into creative work* or *outputs that enhance the enjoyment of creative goods*.

Conceptual Framework

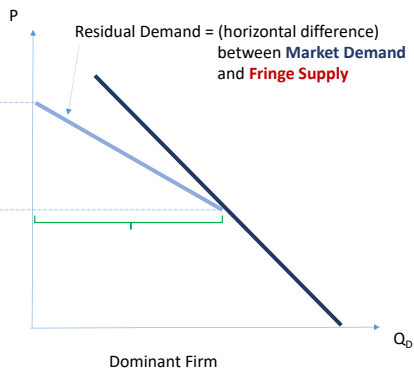
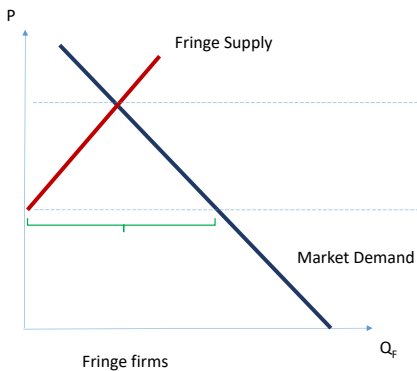
Imagine two countries: **Home** and **Host**

A parent company from **Home** owns a creative work and markets it in **Host**

- Home company is a dominant firm and Host imitators comprises a competitive fringe
- For each creative work, let Q be total output, where $Q = Q_D + Q_F$. Assume Q_D , Q_F are perfect substitutes
- Fringe market is competitive (imagine supply and demand)
- Dominant firm takes the residual market demand
- The smaller the fringe, the greater the profits for the dominant firm
- Hence, stronger copyright protection and enforcement limit fringe supply

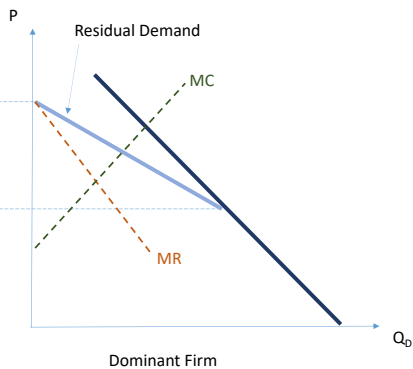
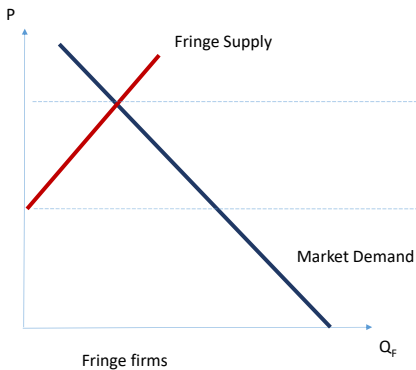
Dominant Firm & Competitive Fringe Model

How to derive residual demand



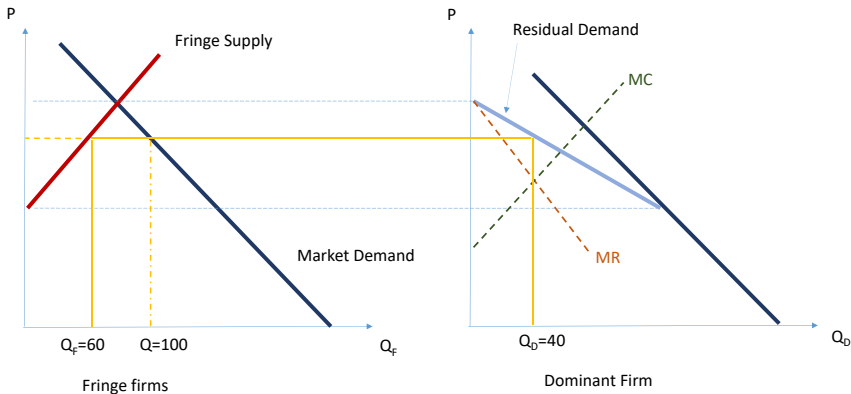
Dominant Firm & Competitive Fringe Model

Maximize profits where $MR = MC$



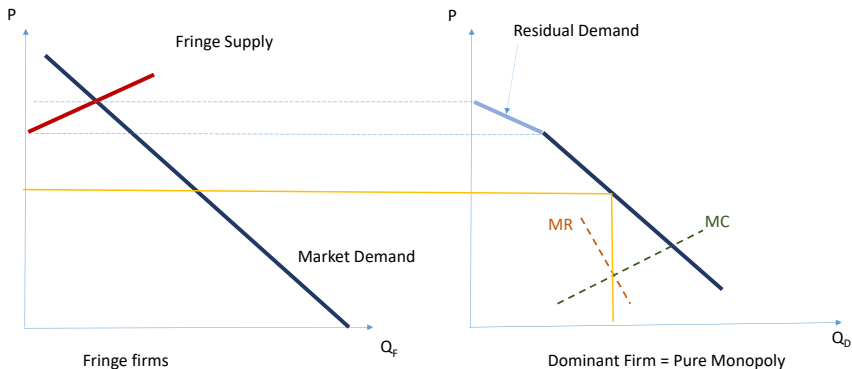
Dominant Firm & Competitive Fringe Model

Output allocation



Dominant Firm & Competitive Fringe Model

Dominant Firm has a big cost advantage



Conceptual Framework ... continued

How copyright protection matters:

- The Home company makes two strategic decisions:
 - I Should it enter the Host Country?
 - II If so, how much should it license (so that affiliates & arms-length parties can produce Q)
- Solve the problem backwards
 - a. In stage II (post-entry), copyright protection has 2 effects: (i) a market expansion effect vs. (ii) market power effect. Both raise profits, but licensing expands if (i) dominates (ii), else simply “fees” increase.
 - ‘Market expansion effect’ matters more if many fringe firms exist
 - Fringe is large if market demand for the creative work is large
 - b. In stage I (pre-entry), let expected stage II profits be Π and the cost of entry be E . If $\Pi > E$, the firm will enter the local market; else, stay out.
 - Entry cost depends on copyright right protection, which raises *transactions costs*, especially when negotiating with *unaffiliated parties*
 - Entry cost depends on adaptation costs, to ensure sufficient product differentiation with local works

Empirical Framework

Hypotheses to be tested, relating to M , market size, and index of copyright strength σ

Hypothesis 1: *Conditional on the firm's knowledge transfer decision, a strengthening of copyright protection in the host country will increase the volume of transfer via licensing relatively more for high-demand creative works.*

Commentary: An increase in σ raises the owner's residual demand (market expansion effect) but increases the owner's ability to exercise market power. A higher M helps offset the market power effect.

Hypothesis 2: *A strengthening of the host's copyright protection will increase the likelihood of knowledge transfer relatively more for high-demand creative works licensed to affiliated parties.*

Commentary: An increase in σ raises the entry cost (e.g., transactions costs), especially for unaffiliated parties who engage in cross-licensing negotiations. A higher M , along with conducting intra-firm transfers, helps offset the higher costs of adaptation.

Estimating Equation

The basic model of knowledge transfer is as follows:

$$\ln(T_{iht}) = \alpha_i + \alpha_h + \alpha_t + \beta_1 \ln C_{ht} + \beta_2 M_j \ln C_{ht} + \ln \mathbf{X}'_{ht} \gamma + \ln \mathbf{Z}'_{it} \varphi + \varepsilon_{iht}$$

- Unit of analysis is the U.S. parent firm i in industry j , licensing its creative work in host country h in year t .
- T_{iht} denotes the volume of licensing
- C_{ht} is the index of effective copyright protection, adjusted for enforcement (i.e., interacted with an index of legal & property rights) in host country h at time t
- M_j is the measure of industry sensitivity to copyright (domestic sales during sample period).
- $\mathbf{X}_{ht}$ is the vector of time-varying host country controls, such as real gross domestic product (GDP), hourly wages relative to the U.S., the quality of economic institutions, and the corporate income tax rate
- $\mathbf{Z}_{it}$ is the vector of time-varying firm controls, such as parent and affiliate sales.
- α_i , α_h , and α_t are the firm, country, and year fixed effects. ε_{iht} is the stochastic error term.
- Estimation methods: Ordinary least squares and two-step *Heckman* selection procedure
- Note: effect of C on T is $\hat{\beta}_1 + \hat{\beta}_2 M_j$

U.S. MNCs

Sixty-nine countries

Micro Data Surveys: 1989 - 2011

Firms in 2 Broad Industries:

- ① Copyright-Related Industries
 - 227 U.S. Parent Companies
- ② Complementary Industries
 - 605 U.S. Parent Companies

Source: BEA

- *Annual Surveys and Benchmark Surveys of USDIA, BE-10 & BE-11*
- *Quarterly Balance of Payment Surveys of USDIA, BE-577*
- *Quarterly Survey of Transactions in Selected Services and Intellectual Property with Foreign Persons, BE-125*
- *Annual Survey of Royalties, Licensing Fees, and Other Receipts and Payments for Intangible Rights between U.S. and Unaffiliated Foreign Persons, BE-93*
- *Quarterly Survey of Transactions between U.S. and Unaffiliated Foreign Persons in Selected Services and in Assets, BE-25*

Industry Coverage

NAICS	Copyright Industries
5111	Newspaper, periodicals, books, and directories
5112	Software publishers
5121	Motion picture and video industries
5122	Sound recording industries
5151	Radio and television broadcasting
5152	Cable and other subscription programming
5161	Internet publishers
5171	Wired telecommunications carriers
5172	Wireless telecommunications (excluding satellite)
5173	Telecommunications re-sellers
5174	Satellite telecommunications
5175	Cable distribution
5179	Other telecommunications
5181	Internet service providers, web search portals
5182	Data processing & hosting
5191	Other information services

NAICS	Complementary Industries
3231	Printing & related support
3341	Computer and peripheral equipment
3342	Communications equipment
3343	Audio and video equipment
3344	Semiconductors, other electronic components
3345	Navigational, measuring, control instruments
3346	Making, reproducing magnetic/optical media
3353	Electrical equipment
3359	Other electrical equipment and components

Measurement of Copyright Strength and Adaptability

Index of Copyright Strength

- Park (2005) and Reynolds (2003), updated to 2011. Source of Information: Statutes, Case Laws, and Commentaries
- Four Components (measuring strength of protection):
 - 1 Duration (Copyrights and Related Rights)
 - 2 Usage (Private Use, Collective Licensing)
 - 3 Enforcement (Injunction, Criminal Penalties, Anti-circumvention)
 - 4 International Agreements/Treaties/Conventions
- Overall Index varies from 0 – 4.
- Note: Index of copyright protection is scaled by a normalized (Fraser Institute) index of legal enforcement, $\theta \in (0, 1)$

Adaptability Index

- Let $f = (f_1, f_2, \dots, f_m)$ be the distribution of creative exports, where $f_z \geq 0$ is the export share of z^{th} good and $\sum_{z=1}^m f_z = 1$
- And $f^* = (f_1^*, f_2^*, \dots, f_m^*)$ is the counterpart for the foreign country.
- The angle θ between f and f^* measures the degree of creative output similarity between the foreign country and the U.S. (home country).

$$\cos(\theta) = \frac{\mathbf{f} \cdot \mathbf{f}^*}{\|\mathbf{f}\| \|\mathbf{f}^*\|} = \frac{\sum_{z=1}^m f_z f_z^*}{\sqrt{\sum_{z=1}^m (f_z)^2} \sqrt{\sum_{z=1}^m (f_z^*)^2}},$$

- Note: $0 \leq \cos(\theta) \leq 1$.

Sample Statistics: Effective Copyright and Adaptability

Index of Effective Copyright Protection

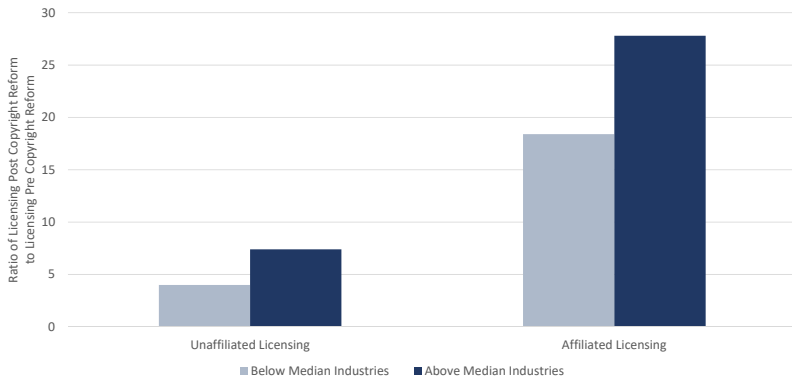
	1994-1999	2000-2005	2006-2011
Mean	1.27	1.56	1.83
Coef of Variation	0.64	0.50	0.40
Skew	0.69	0.36	0.14
Number of Reforming Countries	24	16	6

Degree of Creative Output Similarity w/ U.S.

Region/Industry	Books etc.	Software	Motion Pic.	Sound Rec.	Radio TV
G6	0.98	0.91	0.77	0.86	0.57
Europe (excluding G6)	0.90	0.81	0.84	0.91	0.40
Africa	0.76	0.65	0.64	0.47	0.28
Asia (excluding G6)	0.81	0.77	0.81	0.53	0.32
Latin America & the Caribbean	0.85	0.74	0.68	0.67	0.32
Middle East	0.80	0.67	0.64	0.66	0.33
Oceania	0.89	0.76	0.76	0.84	0.43

Notes: G6 comprises Canada, France, Germany, Italy, Japan, and UK.

Preview of Results



Impacts of Copyrights on Transfers of Creative Works

	Unaff Licen	Aff Licen	Unaff Licen		Aff Licen	
	OLS	OLS	Step 2	Selection	Step 2	Selection
	(1)	(2)	(3)	(4)	(5)	(6)
Copyright prot.	-0.129 (0.289)	-0.607 (0.372)	-0.438*** (0.086)	-0.034 (0.040)	-0.854*** (0.167)	0.161*** (0.062)
Copyright prot. × Indus sens	0.205** (0.094)	0.457*** (0.104)	0.261*** (0.036)	-0.169 *** (0.018)	0.508*** (0.046)	0.284*** (0.027)
GDP	0.816** (0.391)	0.104 (0.851)	2.916*** (0.186)	0.879*** (0.094)	-0.462* (0.269)	1.255*** (0.147)
Host wages	0.332 (0.274)	0.043 (0.468)	0.586*** (0.144)	0.507*** (0.062)	0.254 (0.231)	0.804*** (0.100)
Inc. tax rate	-0.111 (0.080)	0.002 (0.072)	0.145** (0.062)	-0.003 (0.033)	-0.004 (0.080)	-0.071 (0.050)
Inst. quality	0.016 (0.049)	0.025 (0.081)	0.037 (0.027)	0.024* (0.014)	-0.008 (0.036)	0.012 (0.021)
Parent sales	0.014*** (0.004)	-0.009** (0.004)	-0.003 (0.002)	-0.002 (0.001)	-0.000 (0.003)	0.027*** (0.002)
Affiliate sales	0.031*** (0.009)	0.053*** (0.015)				
Adaptation cost				-0.203*** (0.052)		-0.293*** (0.073)
Mills ratio: λ				0.555*** (0.131)		0.097 (0.102)
Constant	-23.734** (11.574)	-0.634 (24.917)	-76.289*** (5.189)	-25.257*** (2.563)	11.167 (7.405)	-35.689*** (4.045)
Observations	7,611	7,611	39,454	39,454	39,454	39,454
Adj R-sq	0.778	0.492				

Heckman selection 2-step used in cols. 3-6. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Impacts of Copyrights on Transfers of Creative Works

Robustness Checks conducted:

- Examined alternative exclusion restrictions, such as an index of linguistic similarity
- Examined alternative conditioning factor, namely domestic R&D
- Eliminated large G6 economies (Canada, France, Germany, Italy, Japan, and U.K.)
- Tested whether licenses were substantive by examining impacts on affiliate R&D and new FDI

Industry Interdependence

Complementary Industry	N	1st	2nd	3rd	4th	5th	6th-10th
Printing & related supplies	38		41.6%	22.1%	5.2%	3.9%	10.4%
Computers & peripherals	99		9.0%	10.1%	5.5%	1.0%	1.5%
Communications equipment	76	1.3%	47.7%	10.5%	1.3%		8.5%
Audio & visual Equipment	15		24.1%	37.9%	17.2%		
Semiconductors	158		4.4%	1.3%	1.3%	0.6%	3.1%
Navigation & control instruments	110		4.1%	7.7%	1.4%	3.2%	6.8%
Magnetic & optical media	6		100%				
Electrical equipment	24		4.3%	8.5%			18.8%
Other elec. equip. & components	79		4.4%		2.6%	1.3%	

Percentage of firms reporting a **copyright-based industry** as 1st, 2nd, 3rd ... largest industry of sale.

N = number of U.S. parent companies. Sample period: 1994-2011.

Cross-Market Impacts

Parents in the Copyright-Based Industries		
	Number of affiliates in the complementary industries	FDI stock (relative to pre-reform) in the complementary industries
Pre-reform	87	1
Post-reform	188	1.5

Parents in the Complementary Industries		
	Number of affiliates in the copyright industries	FDI stock (relative to pre-reform) in the copyright industries
Pre-reform	101	1
Post-reform	356	2.8

Notes: Post-reform FDI Stock is normalized by pre-reform values.

Copyright Reforms and Complementary Industries

	Unaff Licen OLS (1)	Aff Licen OLS (2)	Aff R&D OLS (3)	Unaff Licen Heckman (4)	Aff Licen Heckman (5)	Aff R&D Heckman (6)
Copyright reform	-0.255** (0.099)	-0.231*** (0.087)	-0.417* (0.221)	-0.230*** (0.052)	-0.232*** (0.085)	-0.293* (0.160)
Copyright reform × Indus. sensitivity	0.215*** (0.079)	0.203*** (0.036)	0.331*** (0.124)	0.201*** (0.028)	0.213*** (0.046)	0.277*** (0.087)
Patent rights index	-0.209** (0.085)	0.361** (0.177)	0.057 (0.317)	0.159** (0.079)	1.071*** (0.128)	1.500*** (0.249)
Patent rights index × Product complexity	0.302*** (0.100)	-0.196** (0.077)	0.690*** (0.201)	0.259*** (0.038)	-0.373*** (0.062)	0.109 (0.119)
Constant	-7.129** (3.074)	4.696 (4.791)	2.123 (14.012)	-19.348*** (2.862)	-32.822*** (4.827)	-64.294*** (9.324)
Observations	25,527	25,527	25,527	63,052	63,052	63,052
Adj. R-sq	0.408	0.281	0.373			

Only Step 2 of Heckman reported to conserve space. Patenting cost/GDP used as exclusion restriction.
 Controls incl. country and year fixed effects, GDP, host wages, inc tax rate, inst quality, parent & affil. sales.
 *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Recap

- Copyright-related works comprise a significant share of *knowledge* goods and international technology transfer
- Copyright protection is *positively* associated with U.S. multinational tech transfer in the *copyright-related industries* for firms in industries with *above median sales*; else, negatively associated.
- Copyright protection protects firm profits but also raises the costs of adaptation of creative works. This can reduce the likelihood of technology transfer, particularly with arms-length parties.
- Copyright reform is *positively* associated with tech transfers and investments in *complementary industries* (e.g., computer equipment, electronics, communications) for firms in industries with *above median sales*; else, negatively associated.
- Future work could examine the role of *limitations and exceptions* to copyrights (e.g., Fair Use, Fair Dealing)

Implications for Empirical Study

- ❑ Pose a question related to IPRs (or examine a puzzle/trend)
- ❑ Review prior literature
- ❑ Develop a conceptual framework: build a model, derive testable predictions
- ❑ Collect/create data or find proxies
- ❑ Examine descriptive statistics
- ❑ Perform a controlled study. Think of your empirical framework (causal identification strategy). Check for robustness.
- ❑ Review the quantitative and qualitative findings, and draw inferences.
- ❑ Iterate (as investigation is not a linear process). Examine limitations and potential extensions.

EXERCISES

1. Review Study

- a. How might copyright strength affect licensing (stage I and stage II)?
 - Market power vs. market expansion effect
 - Entry cost, affiliate vs. unaffiliated firms
- b. Why might manufacturing industries be affected by copyright? Provide examples.
 - Computers, communications, infrastructure

2. Sample Data

<https://www.bea.gov/>.

Look for Tools, Interactive Data

Use a spreadsheet like Excel to examine U.S. trade in services (charges for intellectual property) and their correlation with copyright strength abroad.

<http://fs2.american.edu/wgp/www/intl%20copyright.xlsx>

3. Sample Laws

<https://www.wipo.int/portal/en/index.html>.

Look for (Legal) Resources, *WIPO LEX*

Examine Copyright Acts over time for a country, or compare legal texts between countries.